Q&A
COMPUTABILITY & LOGIC

HOMEWORK & EXAMS

Minimal Requirements Homework:

- Minimal 6 (out of 7) assignments
- Deadline re-handins: Thursday May 21, preferably earlier!

Exam:

- Saturday June 13th , from 10:00-13:00 in the new exam hall
  (but do check the exam plan!)
- On **WiseFlow** + **Locked-down Browser** – See Brightspace for pointers to information

Q&A sessions by TAs:

- Watch Brightspace

EXAM FORM

- A 3 hour "written" digital exam consisting of 20 multiple choice questions + 3 open questions
  - Spread evenly over Regular/Context-free, TM/computability, Logic
- The exam will be held on WiseFlow, with the Locked-down Browser.
  - **Download, install and test Locked-down Browser on your computer before the exam**
  - **“Specified means": You can only use the slides through WiseFlow ()**
- There will be two types of multiple-choice questions:
  - One answer: only one answer is correct; you can select one answer
  - Multiple answers: you will have to select ALL correct answers to get maximal points
    (every correct answer gives partial points; a single wrong answer gives 0 points).

EVALUATION

Thanks for all the constructive feedback!! 69/169 (>40%)

Most positively evaluated:

- Structure of the course
- Quality of the handouts
- Engagement teachers / TAs
- Composition of the semester

Constructive comments on:

- Interaction during lectures
- Dense material
- Relevance theory/practice
- Workload ???

	Mean
I rate the overall outcome of the course as:	4.00
The workload on the course is assessed as	3.14
I experience that the course is relevant for my studies on the whole	4.03
The course is well organised	4.55
The student teacher/-s communicated the material in a way that supported my learning - write the student teacher's full name in the comments box	4.64
I have prioritised my work effort in this course compared with other courses this semester:	3.12

CONTENTS OF THE COURSE

Recall basically 4 topics

1. Regular Expressions
   and Finite Automata

2. Context-Free Grammars
   and Push-down Automata

3. Turing Machines and Decidability
   and Complexity

4. Logic: propositional + predicate logic
   Models and Proof system
   Soundness & Completeness

Exam:

- balanced over these parts
- balanced between "skills" and "theory"
  - **Skills**:
    understand the individual models,
    transformations, proof rules
  - **Theory**:
    understand the relationships / classes
    computability/feasibility/completeness
    Closure properties

1. REGULAR LANGUAGES

Alphabet, Words, Languages

Regular Expressions (RE)
- language

Deterministic Finite Automata (DFA)
- accepted language
- (in)-distinguishability
- Pumping Lemma

Non-deterministic Finite Automata (NFA)
- lambda steps
- accepted language

Constructions/skills:
- RE <-> NFA -> DFA
- Minimization DFA
- Pumping Lemma
- Apply Closure Properties
- Modeling/Recognizing

2. CONTEXT-FREE LANGUAGES

Context Free Grammars
- Accepted Language
- Derivation trees, Ambiguity
- Chomsky Normal Form
- CYK parsing

Push-Down Automata
- Accepted Language
- Various acceptance conditions
- Deterministic PDA
- Equivalence Grammar-PDA

Constructions/skills:
- CFG -> PDA
- Construction CNF
- CYK parsing
- Modeling/Recognizing
- Pumping Lemma
- Apply Closure Properties

3. TURING MACHINES

Turing Machine definition
- recognizing languages
- computing functions

Variants
- multiple tapes
- non-deterministic TM

Accepting vs Deciding
- Recursive Languages
- Recursive Enumerable Languages
- Countable Sets

Chomsky Hierarchy
- Unrestricted Grammars
- Equivalences / inclusions

Undecidability
- Halting Problem
- Problem Reductions
- Rice's Theorem
- Post Correspondence Problem
- Decidability Context-Free Languages

Complexity Theory
- P vs NP classes
- Polynomial Reduction

Skills
- modeling/recognizing TMs
- problem reduction
- understand consequences

4. LOGIC, SOUNDNESS & COMPLETENESS

Functions and Equivalence Relations

Different Logics
- propositional logic
- predicate logic
- arithmetic (Peano axioms)

For each logic:
- what are the models
- proof rules
- soundness & completeness status
- decidability status

Skills
- check validity or satisfiability
  (given formula, in some/all models)
- check provability (follow proof rules)
- understand validity / tautology / satisfiability
- understand the relationship between expressiveness and computability
  (differences between prop/pred/arith)

PLAN

Your questions?

Topics?

Test exams?

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